a - PR no.
n - PI, N no.

$\sqrt[n]{a}$

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• Rationalisation factor or Congugate
- If the product of two surds is a rational number then each of the 2 surds is the R.F of other.
↓
Rationalisation factor.

Eg → $\sqrt{2} \times \sqrt{2} = 2$
↓ ↓ ↓
Ir R.F R

• $\sqrt[3]{2} = (2)^{1/3} \times (2)^{2/3} = (2)^{\frac{1}{3}+\frac{2}{3}} = (2)^{3/3} = 2^1 = 2$
Alternate + → −
− → +

• $(2+\sqrt{3}) \cdot (2-\sqrt{3})$
$= 2(2-\sqrt{3}) + \sqrt{3}(2-\sqrt{3})$
$= 4 - 2\sqrt{3} + 2\sqrt{3} - 3$
$= 4-3$
$= 1$

* conjugate (R.F of mixed surd)
$a \pm \sqrt{b} = a \mp \sqrt{b}$
Conjugate

Eg → $(\sqrt{8} - \sqrt{3}) \cdot (\sqrt{8} + \sqrt{3})$
$= (\sqrt{8})^2 - (\sqrt{3})^2$
$= 8 - 3$
$\Rightarrow 5$

Mixed - $(a \pm \sqrt{b}) \cdot (a \mp \sqrt{b})$
(binomial quardatic surd) → $(\sqrt{a} + \sqrt{b})(\sqrt{a} - \sqrt{b})$

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Eg → $(2+\sqrt{3}) \cdot (2-\sqrt{3})$
$= (2)^2 - (\sqrt{3})^2$
$= 4 - 3$
$\Rightarrow 1$ (Rational)

* • conjugate (R.F of binomial quardatic surd)
conjugate
$\sqrt{a} \pm \sqrt{b} \quad \sqrt{a} \mp \sqrt{b}$
invert symbol
conjugate

Eg → $(2\sqrt{3} - 4\sqrt{5}) \cdot (2\sqrt{3} + 4\sqrt{5})$
$\Rightarrow (2\sqrt{3})^2 - (4\sqrt{5})^2$
$\Rightarrow 12 - 80$
$\Rightarrow -68$ Ans

Rationalise the denominator

⇒ fraction $= \frac{N^r}{D^r}$

* If we find irrational number in denominator then we rationalise the denominator by multiplying the denominator with its R.F / conjugate.

Eg → $\frac{1}{\sqrt{3}} \times \frac{\sqrt{3}}{\sqrt{3}} = \frac{\sqrt{3}}{3}$ → Denominator is rational

- $\frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{(2)^2-(\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}$

- $\frac{1}{2\sqrt{7}} \times \frac{2\sqrt{7}}{2\sqrt{7}} = \frac{\sqrt{7}}{14} = \frac{\sqrt{7}}{14}$

- $\frac{\sqrt{50}}{3\sqrt{5}} \times \frac{\sqrt{5}}{\sqrt{5}} = \frac{5\sqrt{10}}{15} = \frac{\sqrt{10}}{3}$

- $\frac{4+\sqrt{3}}{5-\sqrt{5}} \times \frac{5+\sqrt{5}}{5+\sqrt{5}} = \frac{4(5+\sqrt{5}) + \sqrt{3}(5+\sqrt{5})}{(5)^2-(\sqrt{5})^2}$

$= \frac{20+4\sqrt{5}+5\sqrt{3}+\sqrt{15}}{25-5} = \frac{20+4\sqrt{5}+5\sqrt{3}+\sqrt{15}}{20}$

- simplify

- $\frac{4+\sqrt{5}}{4-\sqrt{5}} + \frac{4-\sqrt{5}}{4+\sqrt{5}}$

- $\Rightarrow \frac{4+\sqrt{5}}{4-\sqrt{5}} \times \frac{4+\sqrt{5}}{4+\sqrt{5}} = \frac{(4+\sqrt{5})^2}{(4)^2-(\sqrt{5})^2} = \frac{(4)^2+(\sqrt{5})^2+2(4)(\sqrt{5})}{16-5}$

$= \frac{16+5+8\sqrt{5}}{11} = \frac{21+8\sqrt{5}}{11}$

- $\frac{4-\sqrt{5}}{4+\sqrt{5}} \times \frac{4-\sqrt{5}}{4-\sqrt{5}} = \frac{(4)^2+(\sqrt{5})^2-2(4)(\sqrt{5})}{(4)^2-(\sqrt{5})^2}$

$= \frac{16+5-8\sqrt{5}}{16-5} = \frac{21-8\sqrt{5}}{11}$

Add

$\frac{21+8\sqrt{5}}{11} + \frac{21-8\sqrt{5}}{11}$

$\Rightarrow \frac{42}{11}$

2.) $\frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} + \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

$\Rightarrow \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}} + \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}-\sqrt{2}} \times \frac{\sqrt{3}+\sqrt{2}}{\sqrt{3}+\sqrt{2}}$

$\Rightarrow \frac{3+2-2\sqrt{6}}{3-2} + \frac{3+2+2\sqrt{6}}{3-2}$

$\Rightarrow 5-2\sqrt{6} + 5+2\sqrt{6}$

$\Rightarrow 10$

3) $\frac{\sqrt{5}-1}{\sqrt{5}+1} + \frac{\sqrt{5}+1}{\sqrt{5}-1}$

$\Rightarrow \frac{\sqrt{5}-1}{\sqrt{5}+1} \times \frac{\sqrt{5}-1}{\sqrt{5}-1} + \frac{\sqrt{5}+1}{\sqrt{5}-1} \times \frac{\sqrt{5}+1}{\sqrt{5}+1}$

$\Rightarrow \frac{5+1-2\sqrt{5}}{4} + \frac{5+1+2\sqrt{5}}{4}$

$\Rightarrow \frac{6}{4} + \frac{6}{4}$

$\Rightarrow \frac{\cancel{12}\,3}{\cancel{4}\,1} = 3$

4.) $\frac{6}{2\sqrt{3}-\sqrt{6}} + \frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}} - \frac{4\sqrt{3}}{\sqrt{6}-\sqrt{2}}$

$\frac{6}{2\sqrt{3}-\sqrt{6}} \times \frac{2\sqrt{3}+\sqrt{6}}{2\sqrt{3}+\sqrt{6}}$

$\Rightarrow \frac{6(2\sqrt{3}+\sqrt{6})}{(2\sqrt{3})^2-(\sqrt{6})^2}$

$\Rightarrow \frac{12\sqrt{3}+6\sqrt{6}}{12-6}$

$\Rightarrow \frac{12\sqrt{3}+6\sqrt{6}}{6}$

$\Rightarrow \frac{\cancel{6}(2\sqrt{3}+\sqrt{6})}{\cancel{6}}$

$\Rightarrow 2\sqrt{3}+\sqrt{6}$

$+ \frac{\sqrt{6}}{\sqrt{3}+\sqrt{2}} \times \frac{\sqrt{3}-\sqrt{2}}{\sqrt{3}-\sqrt{2}}$

$\Rightarrow \frac{\sqrt{6}(\sqrt{3}-\sqrt{2})}{(\sqrt{3})^2-(\sqrt{2})}$

$\Rightarrow \frac{\sqrt{18}-\sqrt{12}}{3-2}$

$\Rightarrow \frac{3\sqrt{2}-2\sqrt{3}}{1}$

$\Rightarrow 3\sqrt{2}-2\sqrt{3}$

$- \frac{4\sqrt{3}}{\sqrt{6}-\sqrt{2}} \times \frac{\sqrt{6}+\sqrt{2}}{\sqrt{6}+\sqrt{2}}$

$\Rightarrow \frac{4\sqrt{3}(\sqrt{6}+\sqrt{2})}{(\sqrt{6})^2-(\sqrt{2})^2}$

$\Rightarrow \frac{4\sqrt{18}+4\sqrt{6}}{6-2}$

$\Rightarrow \frac{12\sqrt{2}+4\sqrt{6}}{4}$

$\Rightarrow \frac{\cancel{4}(3\sqrt{2}+\sqrt{6})}{\cancel{4}}$

$\Rightarrow 3\sqrt{2}+\sqrt{6}$

$\Rightarrow (2\sqrt{3}+\sqrt{6}) + (3\sqrt{2}-2\sqrt{3}) - (3\sqrt{2}+\sqrt{6})$

$(\cancel{2\sqrt{3}}+\sqrt{6}) + (3\sqrt{2}-\cancel{2\sqrt{3}}) - (3\sqrt{2}+\sqrt{6})$

$\Rightarrow 2\cancel{\sqrt{3}} + \cancel{\sqrt{6}} + 3\cancel{\sqrt{2}} - 3\cancel{\sqrt{2}} - \cancel{\sqrt{6}} - 2\cancel{\sqrt{3}}$

Q.) Find 'a' and 'b'.

a.) $\frac{3+\sqrt{2}}{3-\sqrt{2}} = a + b\sqrt{2}$

$\Rightarrow \frac{3+\sqrt{2}}{3-\sqrt{2}} \times \frac{3+\sqrt{2}}{3+\sqrt{2}} = a + b\sqrt{2}$

$= \frac{(3)^2+(\sqrt{2})^2+2(3)(\sqrt{2})}{(3)^2-(\sqrt{2})^2} = a + b\sqrt{2}$

$= 9 + 2 + 6\sqrt{2} = a + b\sqrt{2}$

$= \frac{11 + 6\sqrt{2}}{7} = a + b\sqrt{2}$

$\boxed{a = 11/7, \quad b = 6/7}$

b.) $\frac{7+\sqrt{5}}{7-\sqrt{5}} = a + b\sqrt{5}$

$\Rightarrow \frac{7+\sqrt{5}}{7-\sqrt{5}} \times \frac{7+\sqrt{5}}{7+\sqrt{5}} = a + b\sqrt{5}$

$= (7)^2 + (\sqrt{5})^2 + 2(7)(\sqrt{5}) = a + b\sqrt{5}$

$= 49 + 5 + 14\sqrt{5} = a + b\sqrt{5}$

$= \frac{54 + 14\sqrt{5}}{44} = a + b\sqrt{5}$

$\boxed{a = 54/44, \quad b = 14/44}$

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Q. If $a = 2 + \sqrt{3}$, then find $a - \frac{1}{a}$

sol:

given $\rightarrow a = 2 + \sqrt{3}$

then

$$\frac{1}{a} = \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{(2)^2-(\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3}$$

$$\Rightarrow \frac{2-\sqrt{3}}{1} = \boxed{2-\sqrt{3} \Rightarrow \frac{1}{a}}$$

then $a - \frac{1}{a} = ?$

$(2+\sqrt{3}) - (2-\sqrt{3})$

$\Rightarrow \cancel{2} + \sqrt{3} - \cancel{2} + \sqrt{3}$

$\Rightarrow 2\sqrt{3}$

Q. $\frac{\sqrt{2}+\sqrt{3}}{\sqrt{18}-\sqrt{12}} = a - b\sqrt{6} + c\sqrt{3}$, find a, b and c.

sol: $\frac{\sqrt{2}+\sqrt{3}}{\sqrt{18}-\sqrt{12}} \Rightarrow \frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}}$

$$\Rightarrow \frac{\sqrt{2}+\sqrt{3}}{3\sqrt{2}-2\sqrt{3}} \times \frac{3\sqrt{2}+2\sqrt{3}}{3\sqrt{2}+2\sqrt{3}} = a - b\sqrt{6} + c\sqrt{3}$$

$$\Rightarrow \frac{\sqrt{2}(3\sqrt{2}+2\sqrt{3}) + \sqrt{3}(3\sqrt{2}+2\sqrt{3})}{(3\sqrt{2})^2 - (2\sqrt{3})^2}$$

$$= \frac{6 + 2\sqrt{6} + 3\sqrt{6} + 6}{18 - 12} = a - b\sqrt{6} + c\sqrt{3}$$

$$\Rightarrow \frac{12 + 5\sqrt{6}}{6} = a - b\sqrt{6} + c\sqrt{3}$$ (individually divide)

Spiral

$$\boxed{\begin{aligned} a &= 2 \\ b &= -\frac{5}{6} \end{aligned}}$$

$$\underline{\underline{c = 0}}$$

$(3\sqrt{5})^2 = 3\sqrt{5} \times 3\sqrt{5} = 45$

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Q. $\frac{2+3\sqrt{5}}{2-3\sqrt{5}} \times \frac{2+3\sqrt{5}}{2+3\sqrt{5}}$ → $\left[(a+b)^2 = a^2+b^2+2ab \right.$ → $\left. (a+b)(a-b) = a^2-b^2 \right]$

sol:

$\Rightarrow \frac{(2)^2 + (3\sqrt{5})^2 + 2(2)(3\sqrt{5})}{(2)^2 - (3\sqrt{5})^2}$ $[a+b]^2$ $[a^2-b^2]$

$\Rightarrow \frac{4 + 45 + 4 \times 3\sqrt{5}}{4 - 45}$

$\Rightarrow \frac{4 + 45 + 12\sqrt{5}}{-41}$

$\Rightarrow \frac{49 + 12\sqrt{5}}{-41}$ (Minus sign ko upar bhejenge to sign change hojayenge)

$\Rightarrow \frac{-49 - 12\sqrt{5}}{41}$

Q. If $a = 2+\sqrt{3}$ find $a - \frac{1}{a}$

sol:

Given → $2+\sqrt{3}$

so, $\frac{1}{a} = \frac{1}{2+\sqrt{3}}$ (Rationalise) $\left((\sqrt{3})^2 = 3 \right)$

$\Rightarrow \frac{1}{2+\sqrt{3}} \times \frac{2-\sqrt{3}}{2-\sqrt{3}} = \frac{2-\sqrt{3}}{(2)^2-(\sqrt{3})^2} = \frac{2-\sqrt{3}}{4-3} = 2-\sqrt{3}$

$\left[\frac{1}{a} = 2-\sqrt{3} \right]$

$a - \frac{1}{a}$

$\Rightarrow (2+\sqrt{3}) - (2-\sqrt{3}) \Rightarrow \cancel{2}+\sqrt{3} - \cancel{2} + \sqrt{3} = \sqrt{3}+\sqrt{3} = 2\sqrt{3}$

Spiral